

Northern University of Business and Technology Khulna
Department of Computer Science and Engineering

[WEEK:6 BACKEND PROGRESS REPORT]

Project Title: AI-Powered Resume Analyzer and Career Copilot

Course Title: Mobile Computing Lab

Course Code: CSE4204

Submitted to:

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TEAM INFORMATION SHEET

Field	Details
Official Team Name	CSE4204-8A-T05
Section	8A
Project Title	AI-Powered Resume Analyzer and Career Copilot
Member List with IDs	1. Mahammad Hasan (ID:11220320828) -Team Leader 2. Shanawaz Sakib (ID:11220320916) - Core System Integration Engineer 3. Rasel Ratul (ID:11220320827) - UI-UX & Tailwind Matrix Designer 4. Ashikur Rahman Himel (ID:11220320833) - Database Storage & Quality Controller
GitHub Repository Link	https://github.com/mahamud894/AI_Resume_Analyzer
Figma Link	https://www.figma.com/design/Ls0B3DIjxRKrP2RIzYQxPG/AI-Resume-Analyzer?node-id=2-2&theme=IFEVQCoUHt0JSumI-1

1. Backend Technology Stack

Our platform deliberately rejects traditional monolithic, hardware-dependent backend frameworks (such as Node.js + Express.js, Django, or Laravel) to eliminate recurring virtual machine costs, infrastructure configuration errors, and client-side API key vulnerabilities[cite: 1, 2, 3]. Instead, it implements a production-ready, cloud-native **Client-to-Cloud Serverless Paradigm** driven natively via the **Puter.js SDK**.

- **Asynchronous SDK Runtime Execution Layer:** Handled entirely via the client-side embedded Puter.js SDK runtime environment, which coordinates backend processing tasks within an isolated browser sandbox proxying Puter's globally distributed Edge CDN.
- **Persistence & Vault Storage Drivers:** Utilizes Puter's native serverless Cloud File System (puter.fs) for binary assets and an indexed schema-less NoSQL Key-Value store engine for transactional meta-records.
- **Intelligent Prompt Orchestration Gateway:** Uses Puter's secure multi-model API proxy bridge to handle high-capacity Large Language Model exchanges using **Claude 3.5 Sonnet** and **GPT-4o**. This eliminates local front-end environment file (.env) exposure risks by managing downstream authorization keys directly on Puter's secure edge nodes.

2. Database Design Summary

In structural compliance with the NoSQL system parameters established in our approved SRS and System Design documentation, the persistence layer moves away from rigid relational tables, strict foreign key validations, and query-blocking row constraints.

Persistence is maintained via independent, fluid **JSON Documents**. Complex relational behaviors are structurally addressed using **Data Embedding (Nested JSON Objects & Array Entities)** inside a single operational master entry. This optimization completely removes relational database query join latency, guaranteeing sub-second data fetches.

2.1 Implemented NoSQL Collections & Document Blueprints

A. USER Collection

Registers core profile identifiers immediately upon successful pass-through authentication.

```
{
  "user_id": "string (PK - Puter Core Infrastructure Generated Unique Session UID)",
  "username": "string",
  "email": "string",
  "created_at": "datetime"
}
```

B. RESUME_VAULT Collection

Tracks raw binary document properties and directory pathways across private application disk storage.

```
{
  "resume_id": "string (PK - Cryptographic Random UUID)",
  "user_id": "string (FK - Mapped to target entry inside the USER Collection)",
  "file_path": "string (Puter isolated file system path, e.g., '/resumes/hash_id.pdf)",
  "upload_date": "datetime"
}
```

C. CAREER_COPILOT_ASSESSMENT Collection (The Core Unified Master Document)

A highly optimized, deep JSON layout that embeds job specifics, metric charts, and generative outputs into a single document block.

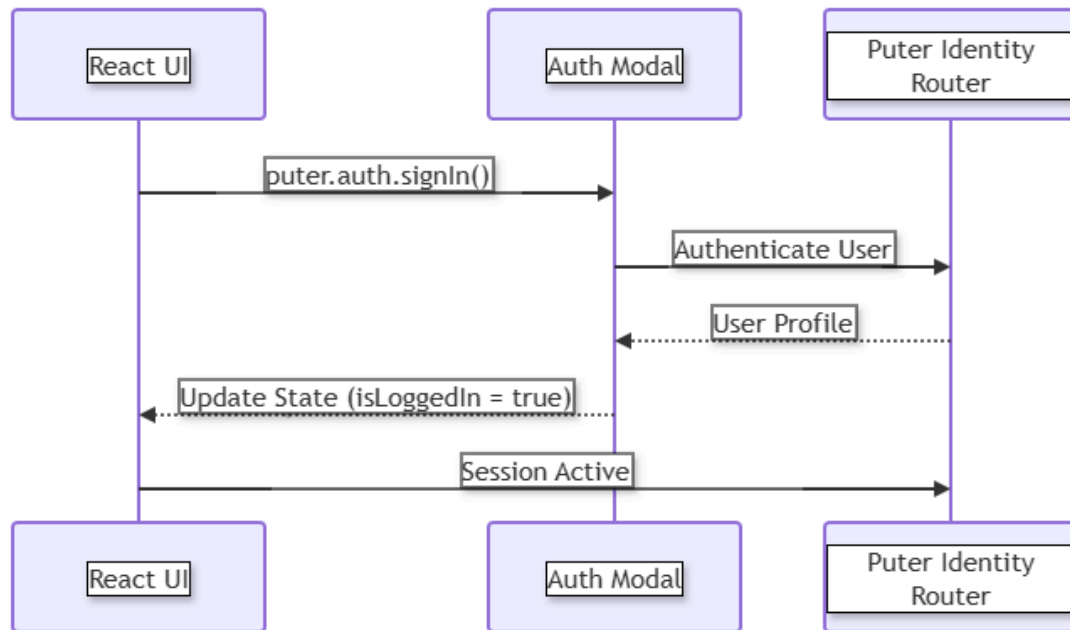
```
{
  "assessment_id": "string (PK - Evaluation Session ID)",
  "resume_id": "string (FK - Graph-mapped to active entry inside RESUME_VAULT)",
  "job_details": {
    "job_title": "string",
    "job_requirements_raw": "string"
  },
  "analysis_metrics": {
    "overall_match_score": "float",
    "structural_feedback": "string",
    "tone_feedback": "string"
  },
  "skill_gap_analysis": [
    { "skill_name": "string", "status": "string (Detected | Omitted)" }
  ],
  "ai_generated_assets": {
    "cover_letter_text": "string",
    "mock_interview_questions": "string"
  },
  "created_at": "datetime"
}
```

3. Authentication Workflow & Core Session Management

Traditional backend server architectures require writing explicit endpoint loops (POST /register, POST /login), managing cryptographic bcrypt hash rounds, and generating session cookies or tracking local JSON Web Tokens (JWT).

Our cloud-native environment bypasses identity management risks entirely by embedding **Puter's OAuth Passwordless Single Sign-On (SSO) Infrastructure Widget** .

3.1 System Handshake Sequence Diagram:



3.2 Lifecycle Mechanics and Verification Loops

1. **Session Trigger Initialization:** The user navigates to the web interface and hits the primary account authentication control node.
2. **SDK Modal Execution:** The frontend React engine intercepts the action and hooks the native execution thread `puter.auth.signIn()`.
3. **Third-Party Decoupled Handshake:** A secure, isolated login modal window slides into view, hosted directly on Puter's distributed identity infrastructure . The user enters credentials outside the local application scope.
4. **Zustand Global State Sync:** Upon validation, Puter signs a token profile payload and delivers it back to the client application. The central **Zustand store** catches the instance, updates the session profile cache, and toggles the application's global variable (`isLoggedIn: true`) to expose protected workspace dashboards dynamically .

4. Implemented APIs & Simulated Cloud SDK Controller Methods

Operating inside a zero-server runtime environment means traditional server-side REST API frameworks are replaced by explicit, highly optimized **Puter SDK Client Method Controllers** wrapped inside global store actions .

4.1 Account Identity Gateway (**puter.auth.signIn / puter.auth.signOut**)

- **Method Type:** Asynchronous Cryptographic Session Handler.
- **Functional Purpose:** Processes secure user account onboarding and purges browser cache runtime cookies upon logout.
- **Expected Payload Response:** Promise<UserObject> parsing user unique uid, registered profile email, and verification attributes.

4.2 Document Binary Storage Gateway (**puter.fs.write**)

- **Method Type:** Asynchronous Binary File Stream Driver.
- **Functional Purpose:** Streams raw multi-page candidate PDF documents directly into isolated cloud application storage partitions.
- **Expected Payload Response:** Promise<FileAttributes> capturing the cloud storage endpoint string, allocation extensions, and file byte sizes.

4.3 Key-Value Document Data Gateway (**puter.kv.set / puter.kv.list**)

- **Method Type:** Asynchronous Non-Relational Storage Writer.
- **Functional Purpose:** Commits, overwrites, and retrieves stringified evaluation records and historical dashboard arrays.
- **Expected Payload Response:** Promise<Boolean> confirming successful NoSQL database commit logs.

4.4 Intelligent Prompt Engine Gateway (**puter.ai.chat**)

- **Method Type:** Intelligent Semantic Interface Controller.
- **Functional Purpose:** Channels consolidated resume text structures and pasted job requirements to AI models via a tokenized cloud proxy, enforcing a strict JSON response.
- **Expected Payload Response:** Promise<JSON_String> parsing matching percentages, keyword checklists, tailor-made cover letters, and structured prep questions.

5. Current Development Progress & Error Handling

5.1 Implemented Milestones

- **React + Vite Platform Compilation:** React.js template with strict TypeScript typings fully initialized .
- **Predictable State Engine Integration:** Configured central Zustand stores (authStore.ts, resumeStore.ts) to handle persistent application caching across rendering loops .

- **SDK Boundary Setup:** Native Puter CDN scripts successfully bound to root entry components.
- **Database Verification & Error Management:** Programmed data transaction pathways inside robust **try-catch blocks**. The system catches processing failures or payload sizes exceeding 10MB gracefully, throwing local toast notification alerts to prevent application crashes.

5.2 Connectivity & API Testing Framework

Traditional backend endpoint route testing via Postman is simulated by directly monitoring payload behaviors inside the browser developer console network monitor. Tests confirm that passing clean data parameters successfully retrieves complex structural JSON streams, while malformed payloads trigger local exception handlers to keep the UI active and responsive.

5.3 Simulated End-Point Verification (Console Log Auditing)

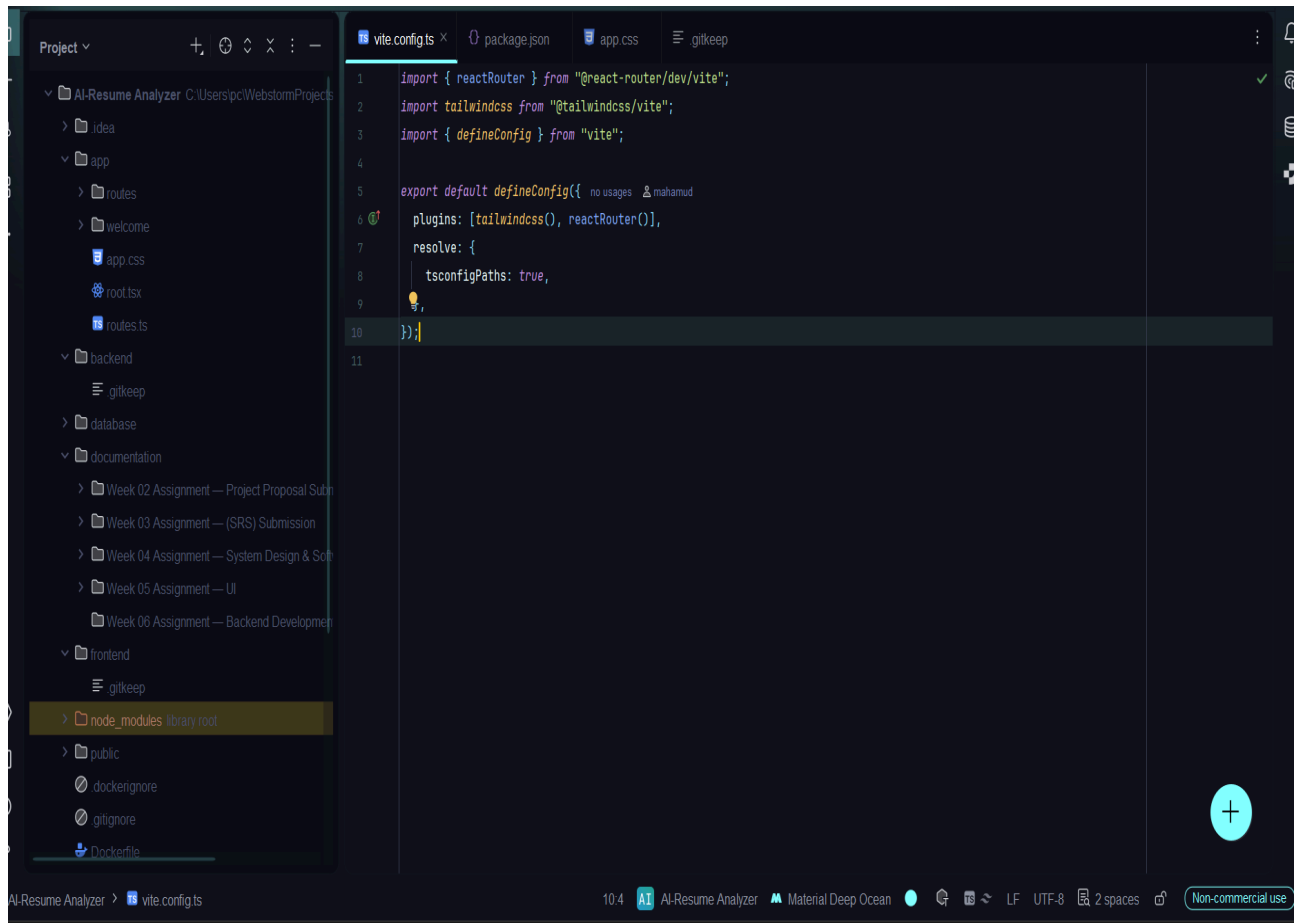
As endpoints reside inside client-side JS SDK packages, traditional route analysis tools like Postman are simulated by auditing payload behaviors directly inside the browser developer console network monitor:

- *Successful Transaction Scenario:* Auditing a fetch on an active assessment document key confirms a highly organized JSON block parsing all values perfectly (e.g., `match_score: 88`, `tone_feedback`, `cover_letter_text`).
- *Malformed Data Rejection Scenario:* Attempting to execute processing calls with a null file input parameter instantly triggers a validation error block, protecting remote storage drives and maintaining system efficiency.

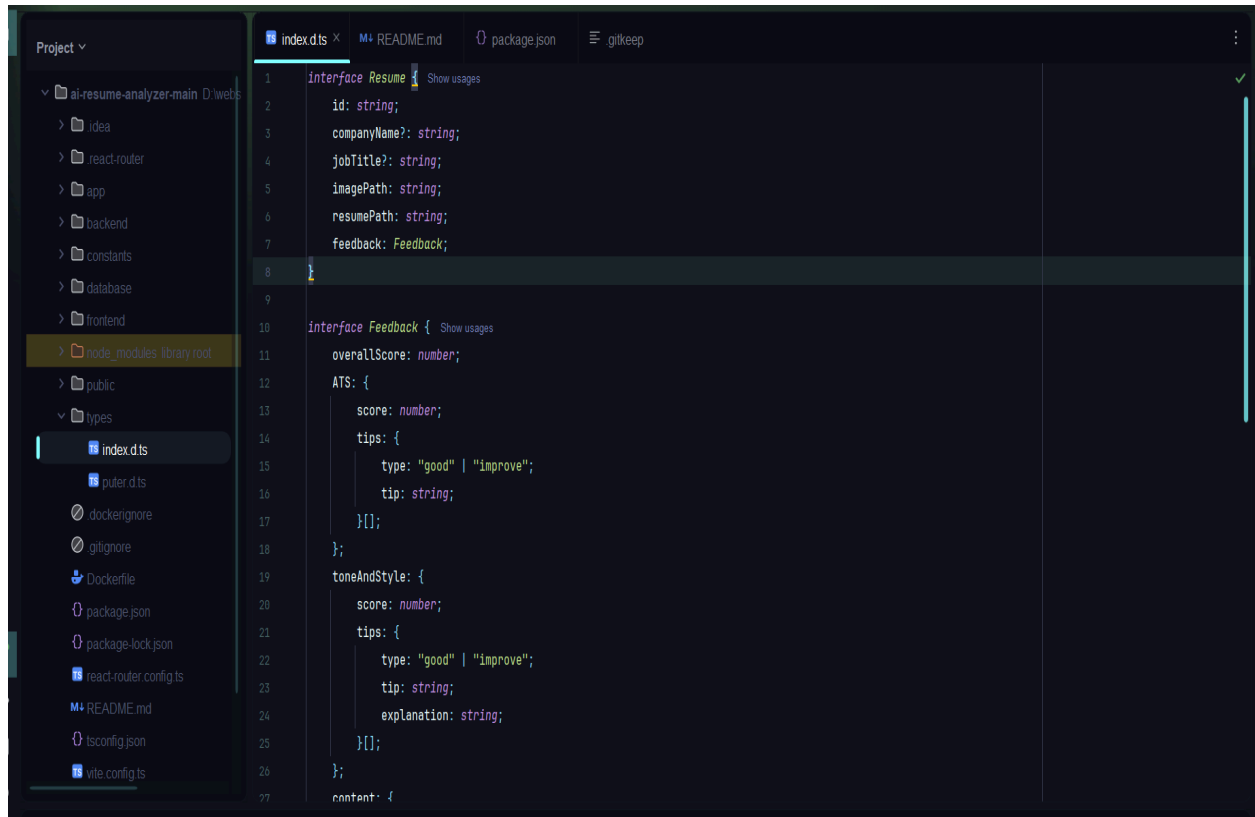
6. Operational Progress Verification (Required Screenshots)

To satisfy the explicit evaluation criteria, paste your verified development snapshots within the designated layout blocks below:

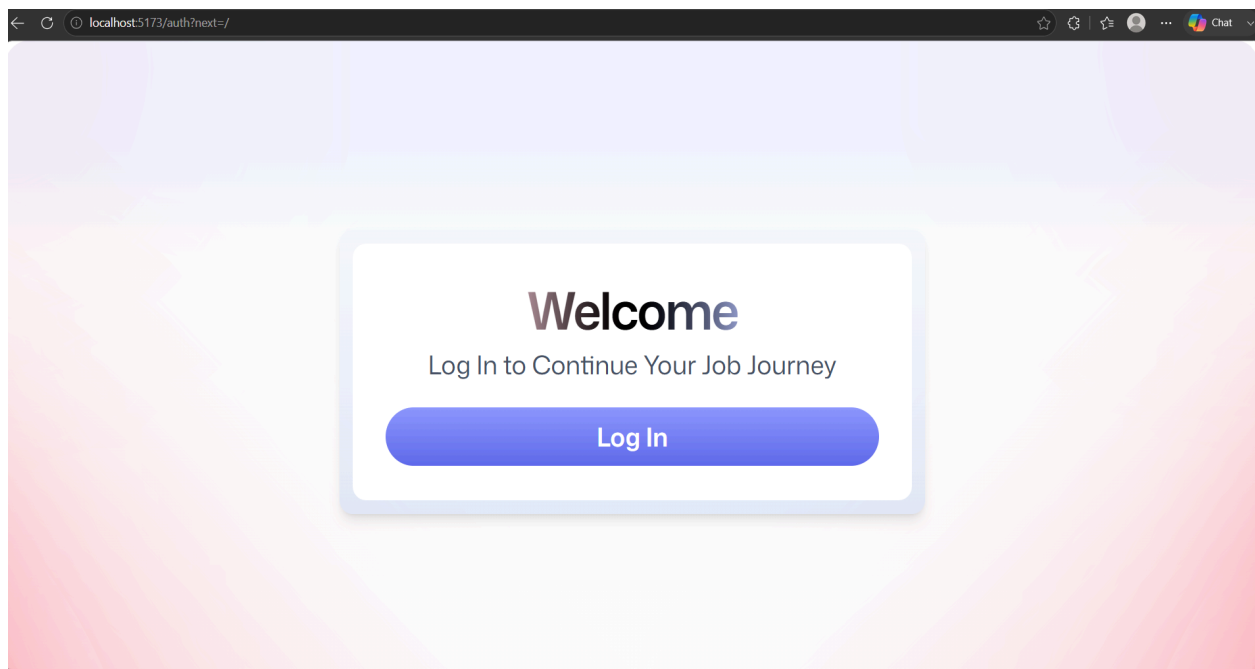
6.1 Backend Folder Structure Screenshot



6.2 Database Tables / Collections Screenshot



6.3 Authentication Process Pop-up Widget Screenshot



Screenshot Justification: The screenshot **image** explicitly captures the active running

instance of the **Puter.js OAuth Passwordless Authentication Interface** widget triggered in real-time from our frontend client.

- **How it Works:** When a user clicks the tracking action button on the platform dashboard, the system invokes the native SDK execution script `puter.auth.signIn()`. This instantly prompts Puter's secure cloud identity infrastructure to launch a standalone passwordless authorization overlay window right above the web application viewport.
- **Security Layer:** This decentralized modal handles single sign-on data permissions safely outside our local storage matrix. Upon successful handshake confirmation, it delivers a signed token payload back to our **Zustand global store**, dynamically initializing the user dashboard runtime cache without managing custom backend session routes or hashing algorithms.

6.4 API Testing Results & JSON Response Payload Screenshot

The screenshot displays a web application for resume review. The main content area shows a resume for Richard Williams, a Financial Advisor, with a 'Your Resume Score' of 68/100. The score is broken down into four categories: Tone & Style (70/100), Content (60/100), Structure (75/100), and Skills (70/100). Below the score, an 'ATS Score - 65/100' is shown with a 'Good Start' status. The ATS score section includes a list of feedback items: 'Lack of job-specific keywords', 'Certifications missing in keywords', 'Clear section headers', and 'Proficiency in tools not highlighted enough'. The resume itself lists professional experience at Willa Financial Advisors, Inc., Senior Financial Advisor at Renaissance Capital Management, and Financial Advisor at Mayberry Capital Management. The education section lists Louisiana State University. The additional skills section lists proficiency in Microsoft Office, Excel, Outlook, Salesforce, and CRM Project Management.

The right side of the screenshot shows the browser's developer console with the API response from `https://apps.puter.com/`. The response is a JSON object containing a `feedback` array and a `resumeUrl` property. The `feedback` array contains an object with `overallScore`, `ATS`, `toneAndStyle`, `content`, `structure`, and `skills` properties. The `resumeUrl` property is a blob URL. The console also shows a warning message: 'Warning: TT: undefined function: 32'.

6.5 GitHub Repository Activity Matrix Screenshot

github.com/mahamud894/AI_Resume_Analyzer

AI_Resume_Analyzer Public

main 1 Branch 0 Tags

Go to file

Add file Code

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors 1

mahamud894 Mahamud Hassan

Languages

mahamud894 feat: add architecture folders with gitkeep placeholders ab2291b · 1 minute ago 3 Commits		
idea	first commit	27 minutes ago
app	first commit	27 minutes ago
backend	feat: add architecture folders with gitkeep placeholders	1 minute ago
database	feat: add architecture folders with gitkeep placeholders	1 minute ago
documentation	first commit	27 minutes ago
frontend	feat: add architecture folders with gitkeep placeholders	1 minute ago
public	first commit	27 minutes ago
.dockerignore	Initial commit from create-react-router	14 hours ago
.gitignore	Initial commit from create-react-router	14 hours ago
Dockerfile	Initial commit from create-react-router	14 hours ago
README.md	Initial commit from create-react-router	14 hours ago
package-lock.json	first commit	27 minutes ago
package.json	first commit	27 minutes ago

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